

Quantum Field Theory

Exercise 5.1

$$I = \int d^4 k \frac{1}{k^2} \frac{1}{(k+p)^2 - m^2}$$

Feynman parametrization:

$$\frac{1}{AB} = \int_0^1 dx \frac{1}{(Ax + B(1-x))^2}$$

Let's use that for I :

$$I = \int d^4 k \int_0^1 dx \frac{1}{[(k+p)^2 - m^2]x + k^2(1-x)^2}$$

$$I = \int d^4 k \int dx \left[\underline{k^2} + p^2 x + 2kp_x - m^2 x + k^2 - \underline{k^2 x} \right]^{-2}$$

$$= \int d^4 k \int dx \left[p^2 x + 2kp_x - m^2 x + k^2 \right]^{-2} = \int d^4 k \int dx \left[\underbrace{(k+px)^2}_{k\text{-dependent part}} - p^2 x^2 - m^2 x + p^2 x \right]^{-2}$$

We have isolated the k -dependent part with other terms (both are still dependent on x)

→ Let's shift the integration variable: $l = k + px \Rightarrow \frac{dk}{dx} = 1 \Rightarrow dl = dk$
and let's just write $\Delta^2 \equiv p^2 x - m^2 x - p^2 x^2$

$$\Rightarrow I = \int dx \int d^4 l (l^2 - \Delta^2)^{-2}$$

We now Wick-rotate to get from Minkowski space to Euclidean space with the help of Cauchy's theorem: variable change from $l \rightarrow l_E$ with $l^0 = il_E^0$, $l^i = l_E^i$

$$\Rightarrow \left[\int_{-\infty}^{+\infty} + \int_{+i\infty}^{-i\infty} \right] dl^0 (l^2 - \Delta^2)^{-2} = 0 \quad \parallel \text{Cauchy's theorem}$$

$$\Rightarrow \int_{-\infty}^{+\infty} dl^0 (l^2 - \Delta^2)^{-2} = - \int_{+i\infty}^{-i\infty} dl^0 (l^2 - \Delta^2)^{-2}$$

$$= -i \int_{-\infty}^{+\infty} dl_E^0 (-l_E^0{}^2 - l_E^i{}^2 - \Delta^2)^{-2}$$

$$= i \int_{-\infty}^{+\infty} dt (-t^2 - r^2 - \Delta^2)^{-2}$$

$$\int_{-\infty}^{+\infty} dl^0 (l^2 - \Delta^2)^{-2} = i \int_{-\infty}^{+\infty} dt [t^2 + r^2 + \Delta^2]^{-2}$$

Now here we use the

$$l^0 = il_E^0, \quad l^i = l_E^i \Rightarrow dl^0 = i dl_E^0$$

limit of integration

$$-i\infty = il_E^0 \Rightarrow l_E^0 \Rightarrow -\infty$$

$$i\infty = il_E^0 \Rightarrow l_E^0 \Rightarrow \infty$$

let's write $l_E^0 = t$

$$\text{and } l_E^i = r^i \Rightarrow t^2 + r^2 = l_E^2$$

$$(-1)^{-2} = 1$$

So l_E is Euclidean 4-dimensional vector

Minkowski

Now can write $\int d^4 l$ integral in terms of Euclidean integral $d^4 l_E$

$$\Rightarrow \int d^4 l (l^2 - \Delta^2)^{-2} = i \int d^4 l_E [l_E^2 + \Delta^2]^{-2}$$

To evaluate this integral we can use the spherical coordinates:

$$\int d^4 l_E = \int d\Omega_4 \int |l_E|^3 d|l_E| \quad \text{with } \int d\Omega_4 = 2\pi^2$$

$$\begin{aligned} \Rightarrow I &= i 2\pi^2 \int dx \int d|l_E| |l_E|^3 (|l_E|^2 + \Delta^2)^{-2} \\ &= 2\pi^2 i \int dx \int_0^\infty d|l_E| \frac{|l_E|^3}{(|l_E|^2 + \Delta^2)^2} \\ &= 2\pi^2 i \int dx \int_0^\infty d|l_E| \frac{d}{d|l_E|} \left\{ -\frac{|l_E|^2}{|l_E|^2 + \Delta^2} + \ln(|l_E|^2 + \Delta^2) \right\} \\ &= 2\pi^2 i \int dx \left[-\frac{|l_E|^2}{|l_E|^2 + \Delta^2} + \ln(|l_E|^2 + \Delta^2) \right] \Bigg|_{|l_E|=0}^{|l_E|=\infty} \end{aligned}$$

$$\frac{d}{d\xi} \frac{\xi^2}{(\xi^2 + \Delta^2)} = -\frac{2\xi^3}{(\xi^2 + \Delta^2)^2} + \frac{2\xi}{\xi^2 + \Delta^2}$$

$$\frac{d}{d\xi} \ln(\xi^2 + \Delta^2) = \frac{2\xi}{\xi^2 + \Delta^2}$$

$$\text{So } \frac{1}{2} \frac{d}{d\xi} \left[-\frac{\xi^2}{\xi^2 + \Delta^2} + \ln(\xi^2 + \Delta^2) \right] = \frac{\xi^3}{(\xi^2 + \Delta^2)^2}$$

This has divergence at $|l_E| \rightarrow \infty$
so let's write in terms of $|l_E|$ as upper limit

$\uparrow$ diverges as $|l_E| \rightarrow \infty$

Since we encounter divergence at $|l_E| \rightarrow \infty$ limit we just calculated the integral without taking the limit to ∞ (since it diverges). We will just write this as the integral but it should be noted that there is still limit $|l_E| \rightarrow \infty$ to be taken. Essentially we have introduced a cut-off at some $|l_E|$.

Now let's regularize the encountered divergence via Pauli-Villars regularization:

This amounts to $\frac{1}{k^2} \rightarrow \frac{1}{k^2} - \frac{1}{k^2 - \Lambda^2}$, where Λ is large

Now we have a integral $I_{pv} = \int d^4 k \left[\frac{1}{k^2} - \frac{1}{k^2 - \Lambda^2} \right] \frac{1}{(k+p)^2 - m^2}$

$$I_{pv} = \underbrace{\int d^4 k \frac{1}{k^2} \frac{1}{(k+p)^2 - m^2}}_{= I} - \underbrace{\int d^4 k \frac{1}{k^2 - \Lambda^2} \frac{1}{(k+p)^2 - m^2}}_{= J}$$

We have already calculated I so let's now calculate J

$$J = \int d^4k \frac{1}{k^2 - \Delta^2} \frac{1}{(k+p)^2 - m^2}$$

// Feynman parametrization: $\frac{1}{AB} = \int_0^1 dx \frac{1}{(Ax + (1-x)B)^2}$
 in our case: $A = (k+p)^2 - m^2$ and $B = k^2 - \Delta^2$

$$= \int dx \int d^4k \left[(k+p)^2 x - m^2 x + k^2 - \Delta^2 - k^2 x + \Delta^2 x \right]^{-2} = \int dx \int d^4k \left[k^2 x + p^2 x + 2kpx - m^2 x - k^2 x + \Delta^2 x - \Delta^2 \right]^{-2}$$

$$(k+px)^2 = 2kpx + k^2 + p^2 x^2 \rightarrow p^2 x + 2kpx - m^2 x + \Delta^2 x - \Delta^2 = (k+px)^2 - \underbrace{p^2 x^2 + p^2 x - m^2 x + \Delta^2 x - \Delta^2}_{= \Delta_\Delta^2 \text{ (doesn't depend on } k)}$$

$$\rightarrow J = \int dx \int d^4k \left[(k+px)^2 - \Delta_\Delta^2 \right]^{-2} \quad // \text{ let's do a shift: } l = k+px \rightarrow dk = dl$$

$$\Rightarrow J = \int dx \int d^4l \left(l^2 - \Delta_\Delta^2 \right)^{-2}, \text{ Now we Wick-rotate like previously with}$$

$$l^0 = i l_E^0, \quad l^i = l_E^i \Rightarrow dl^0 = i dl_E^0 \text{ to get from Minkowski space to Euclidean space.}$$

$$\Rightarrow \int_{-\infty}^{\infty} dl^0 (l^2 - \Delta_\Delta^2)^{-2} = i \int_{-\infty}^{\infty} dl_E^0 \left[l_E^2 + \Delta_\Delta^2 \right]^{-2} \quad // \text{ with } l_E^2 = l_E^1 + l_E^2$$

This gives us $J = i \int dx \int d^4l_E \left[l_E^2 + \Delta_\Delta^2 \right]^{-2}$ // We can use the spherical coordinates with $\int d\Omega_4 = 2\pi^2$

$$J = 2\pi^2 i \int dx \int dl_E l \frac{l_E^3}{(l_E^2 + \Delta_\Delta^2)^2}$$

// This is exactly same integral as I but with Δ_Δ instead of Δ which does not effect the $dl_E l$ integral.

$$\rightarrow J = 2\pi^2 i \int dx \left\{ -\frac{l_E^2}{l_E^2 + \Delta_\Delta^2} + \ln(l_E^2 + \Delta_\Delta^2) \right\} \Bigg|_{l_E=0}^{l_E=\infty}$$

We once again only write/calculate the $|l_E|=0$ limit and leave $|l_E|=\infty$ diverging limit implicitly and get

$$J = -2\pi^2 i \int dx \left[\ln(\Delta_\Delta^2) + \frac{l_E^2}{l_E^2 + \Delta_\Delta^2} + \ln(l_E^2 + \Delta_\Delta^2) \right]$$

Now we can subtract this from I to get I_{pr}

$$I_{pr} = I - J = -2\pi^2 i \int dx \left\{ \ln(\Delta^2) - \ln(\Delta_\Delta^2) + \frac{l_E^2}{l_E^2 + \Delta^2} - \frac{l_E^2}{l_E^2 + \Delta_\Delta^2} + \underbrace{\ln[l_E^2 + \Delta^2] - \ln[l_E^2 + \Delta_\Delta^2]}_{= \ln \left[\frac{l_E^2 + \Delta^2}{l_E^2 + \Delta_\Delta^2} \right]} \right\}$$

Since there is implicit limit of $|l_E| \rightarrow \infty$ we now take the limit

and the terms of form $\frac{l_E^2 \pm u}{l_E^2 \pm v} \rightarrow 1$ as $|l_E| \rightarrow \infty$ and $\ln(1) = 0$

$$\Rightarrow I_{pr} = -2\pi^2 i \int dx \ln \left[\frac{\Delta^2}{\Delta_\Delta^2} \right] = \underline{\underline{2\pi^2 i \int dx \ln \left[\frac{\Delta_\Delta^2}{\Delta^2} \right]}}$$

This is the logarithmic divergence we were looking for.

Exercise 5.2

Compute

$$\int d\Omega_k \left[\frac{2\vec{p} \cdot \vec{p}'}{(\vec{p} \cdot \vec{k})(\vec{p}' \cdot \vec{k})} - \frac{m^2}{(\vec{p} \cdot \vec{k})^2} - \frac{m^2}{(\vec{p}' \cdot \vec{k})^2} \right]$$

Let's work in a frame where $\vec{p}' = -\vec{p} \Rightarrow p^2 = p'^2 = m^2$ since $\vec{p}'$ and $\vec{p}$ is momentum of same particle (electron). $q^2 = (\vec{p} - \vec{p}')^2 \Rightarrow -(2\vec{p})^2 = -4|\vec{p}|^2$ in the chosen frame.

$$\vec{p} \cdot \vec{k} = E \cdot k^0 - \vec{p} \cdot \vec{k}$$

$$E = \sqrt{m^2 + |\vec{p}|^2} = |\vec{p}| \sqrt{\frac{m^2}{|\vec{p}|^2} + 1}$$

$$\beta = \frac{m^2}{(-q^2)} \quad \frac{m^2}{|\vec{p}|^2} = \frac{-4m^2}{-4|\vec{p}|^2} = \frac{4m^2}{-q^2} = 4\beta$$

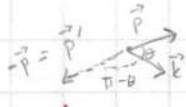
$$E = |\vec{p}| \sqrt{4\beta + 1} = |\vec{p}| \sqrt{1 + 4\beta} \quad \text{and because } k^2 = 0 \text{ we have } k^0 = |\vec{k}|$$

$$\Rightarrow E \cdot k^0 = |\vec{p}| |\vec{k}| \sqrt{1 + 4\beta} \quad \text{Now let's define the angle between}$$

$\vec{p}$ and $\vec{k}$ to same as the angle θ in Ω_k by setting the coordinate to have origin at the necessary point. With this we can write

$$\vec{p} \cdot \vec{k} = |\vec{p}| |\vec{k}| \cos \theta \quad \text{and} \quad \vec{p}' \cdot \vec{k} = |\vec{p}'| |\vec{k}| \cos(\theta') = |\vec{p}| |\vec{k}| \cos(\pi - \theta) = -|\vec{p}| |\vec{k}| \cos \theta$$

Since



$$\Rightarrow \vec{p} \cdot \vec{k} = |\vec{p}| |\vec{k}| (\sqrt{1 + 4\beta} - \cos \theta) \quad \text{and}$$

$$\vec{p}' \cdot \vec{k} = |\vec{p}| |\vec{k}| (\sqrt{1 + 4\beta} + \cos \theta)$$

$$\text{And finally } \vec{p} \cdot \vec{p}' = E^2 + |\vec{p}|^2 = |\vec{p}|^2 [1 + 4\beta + 1] = |\vec{p}|^2 (2 + 4\beta)$$

Let's Now input these to the integral

$$I \equiv \int d\Omega_k \left(\frac{2\vec{p} \cdot \vec{p}'}{(\vec{p} \cdot \vec{k})(\vec{p}' \cdot \vec{k})} - \frac{m^2}{(\vec{p} \cdot \vec{k})^2} - \frac{m^2}{(\vec{p}' \cdot \vec{k})^2} \right) = \int d\Omega_k \left[\frac{|\vec{p}|^2 (2 + 4\beta) \cdot 2}{|\vec{p}| |\vec{k}| (\sqrt{1 + 4\beta} - \cos \theta) |\vec{p}| |\vec{k}| (\sqrt{1 + 4\beta} + \cos \theta)} - \frac{m^2}{|\vec{p}|^2 |\vec{k}|^2 (\sqrt{1 + 4\beta} - \cos \theta)^2} - \frac{m^2}{|\vec{p}|^2 |\vec{k}|^2 (\sqrt{1 + 4\beta} + \cos \theta)^2} \right]$$

$$\Rightarrow I = \frac{2\pi}{|\vec{k}|^2} \int d\cos \theta \left[\frac{4(1 + 2\beta)}{R^2 - \cos^2 \theta} - \frac{4\beta}{(R - \cos \theta)^2} - \frac{4\beta}{(R + \cos \theta)^2} \right]$$

Let's first solve the first term in the integral by writing $u \equiv \cos \theta$ for convenience

$$\int \frac{1}{R^2 - u^2} du = \frac{1}{(R - u)(R + u)} \quad \text{we can factorize this to be of form } \frac{a}{R - u} + \frac{b}{R + u} \quad \text{for some } a \text{ and } b$$

$$x^2 - y^2 = (x + y)(x - y)$$

let's define

$$R \equiv \sqrt{1 + 4\beta} \quad \text{for convenience}$$

$$\Omega_k = d\cos \theta d\phi \quad \text{and } \int d\phi = 2\pi$$

$$\text{also } \frac{m^2}{|\vec{p}|^2} = 4\beta$$

The limit for $\cos \theta$ is from -1 to 1

$$\frac{a}{(R-u)} + \frac{b}{(R+u)} = \frac{aR+au+bR-bu}{R^2-u^2} = \frac{u(a-b) + R(a+b)}{R^2-u^2}$$

This is equal to $\frac{1}{R^2-u^2}$ with $a=b=\frac{1}{2R}$

$$\begin{aligned} \Rightarrow \int_{-1}^1 \frac{1}{R^2-u^2} du &= \frac{1}{2R} \int_{-1}^1 \frac{1}{R-u} + \frac{1}{R+u} du = \frac{1}{2R} \left[\underbrace{-\ln[R-u] + \ln[R+u]}_{=\ln\left(\frac{R+u}{R-u}\right)} \right] \Bigg|_{u=-1}^{u=1} \\ &= \frac{1}{2R} \left[\ln\left[\frac{R+1}{R-1}\right] - \ln\left[\frac{R-1}{R+1}\right] \right] = \frac{1}{R} \ln\left[\frac{R+1}{R-1}\right] \\ &= \frac{1}{\sqrt{1+4\beta}} \ln\left(\frac{\sqrt{1+4\beta}+1}{\sqrt{1+4\beta}-1}\right) \end{aligned}$$

And the second term is integrated as (again we set $u = \cos\theta$ for convenience $\rightarrow du = -\sin\theta d\theta$)

$$\begin{aligned} \int_{-1}^1 \frac{1}{(R-u)^2} + \frac{1}{(R+u)^2} &= \frac{1}{R-u} - \frac{1}{R+u} \Bigg|_{u=-1}^{u=1} \quad \parallel \frac{d}{dx} \frac{1}{R-x} = -1 \cdot \frac{-1}{(R-x)^2} = \frac{1}{(R-x)^2} \\ &= \frac{1}{R-1} - \frac{1}{R+1} - \frac{1}{R+1} + \frac{1}{R-1} = \frac{2}{R-1} - \frac{2}{R+1} \\ &= \frac{2(R+1) - 2(R-1)}{R^2-1} = \frac{4}{R^2-1} = \frac{4}{1+4\beta-1} = \frac{1}{\beta} \end{aligned}$$

Now we put these results together to get

$$I = \frac{2\pi}{|K|^2} \cdot \int d\cos\theta \quad 4(1+2\beta) \frac{1}{R^2-\cos^2\theta} - \frac{4\beta}{\beta} \left[\frac{1}{(R-\cos\theta)^2} + \frac{1}{(R+\cos\theta)^2} \right]$$

$$= \frac{2\pi}{|K|^2} \left[\frac{4(1+2\beta)}{\sqrt{1+4\beta}} \ln\left(\frac{\sqrt{1+4\beta}+1}{\sqrt{1+4\beta}-1}\right) - \frac{4\beta}{\beta} \right]$$

$$I = \frac{8\pi}{|K|^2} \left[\frac{1+2\beta}{\sqrt{1+4\beta}} \ln\left(\frac{\sqrt{1+4\beta}+1}{\sqrt{1+4\beta}-1}\right) - 1 \right]$$

We have computed the angular integral!!

Exercise 53

Calculate: $I \equiv \int dx dy dz \delta(1-x-y-z) \frac{1}{-xyq^2 + (1-z)^2 m^2 + \mu^2}$

Integrate over dx : $I = \int_0^1 dz \int_0^{1-z} dy \frac{1}{-(1-y-z) y q^2 + (1-z)^2 m^2 + \mu^2}$

Change of variables: $y = (1-z)\xi \Rightarrow dy = (1-z)d\xi$ and $w = (1-z)$

$$\Rightarrow \frac{dw^2}{dz} = -2(1-z) \Rightarrow dz = \frac{-1}{2(1-z)} d(w^2) \Rightarrow y = w\xi, \Rightarrow 1-y-z = w-w\xi$$

$$\Rightarrow (1-y-z)y = \xi(1-\xi)w^2 \quad \text{and} \quad (1-z)^2 = w^2$$

$$\Rightarrow I = \int_0^1 d\xi \int_1^0 \frac{-1}{2} \cdot \frac{d(w^2)}{-(\xi(1-\xi))w^2 + w^2 m^2 + \mu^2} \quad \parallel \quad \frac{m^2}{-q^2} = \beta \quad \text{and let's define } R \equiv \frac{\mu^2}{-q^2}$$

$$= \frac{1}{2} \int_0^1 d\xi \int_0^1 \frac{1}{-q^2} \frac{d(w^2)}{[\xi(1-\xi) + \beta]w^2 + R^2} \quad \parallel \quad \frac{d}{dx} \ln(ax+b) = \frac{a}{ax+b}$$

$$= \frac{-1}{2q^2} \int_0^1 d\xi \frac{1}{\xi(1-\xi) + \beta} \left(\ln[\xi(1-\xi) + \beta + R^2] - \ln[R^2] \right)$$

As $R^2 \rightarrow 0$ the term $\frac{1}{\xi(1-\xi) + \beta} \ln[\xi(1-\xi) + \beta + R^2]$ is very small

Since the value of $\frac{1}{\xi(1-\xi) + \beta} \ln[\xi(1-\xi) + \beta]$ is very small since $0 < \xi < 1$

and as $R \rightarrow 0$

the another term diverges $\Rightarrow$ the $\ln[R^2]$ is dominant and $\Rightarrow$ than $\ln[\xi(1-\xi) + \beta + R^2]$ term so we can approximate

$$I \approx - \frac{\ln[R^2]}{2q^2} \int_0^1 d\xi \frac{1}{\xi(1-\xi) + \beta}$$

Let's look for root of polynomial $\xi(1-\xi) + \beta$ to factorize the integrand

$$\xi - \xi^2 + \beta \quad \text{has root of} \quad \xi_0 = \frac{-1 \pm \sqrt{1+4\beta}}{-2}$$

$$\Rightarrow \xi - \xi^2 + \beta = - \left(\xi - \frac{1}{2} - \frac{\sqrt{1+4\beta}}{2} \right) \left(\xi - \frac{1}{2} + \frac{\sqrt{1+4\beta}}{2} \right) \quad \parallel \quad \text{let's write } \frac{\sqrt{1+4\beta}}{2} \equiv k$$

let's also do change of variables: $u = (\xi - \frac{1}{2}) \Rightarrow d\xi = du$

$$\Rightarrow I = - \frac{\ln(R^2)}{2q^2} \int_{-\frac{1}{2}}^{\frac{1}{2}} du \frac{1}{(u-k)(u+k)} = - \frac{\ln(R^2)}{2q^2} \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{du}{(k-u)(k+u)} \quad \parallel \quad \frac{1}{k-u} - \frac{1}{k+u} = \frac{1}{2k} \left[\frac{1}{k+u} + \frac{1}{k-u} \right]$$

$$I = - \frac{\ln(R^2)}{2q^2} \frac{1}{2k} \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{k+u} + \frac{1}{k-u} = - \frac{1}{2q^2} \frac{1}{2k} \left[\ln(k+u) - \ln(k-u) \right]_{u=-\frac{1}{2}}^{u=\frac{1}{2}}$$

$$= \frac{-\ln(R^2)}{2q^2 k} \ln \left[\frac{k+\frac{1}{2}}{k-\frac{1}{2}} \right] = \frac{1}{2q^2} \frac{-2}{\sqrt{1+4\beta}} \left[\frac{\frac{1}{2}(\sqrt{1+4\beta} + 1)}{\frac{1}{2}(\sqrt{1+4\beta} - 1)} \right] \quad \parallel \quad R^2 = \frac{-q^2}{\mu^2}$$

$$= \frac{1}{2} \frac{1}{q^2 \sqrt{1+4\beta}} \ln \left(\frac{-q^2}{\mu^2} \right) \left[-2 \ln \frac{\sqrt{1+4\beta} + 1}{\sqrt{1+4\beta} - 1} \right]$$

We have calculated the integral ???